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SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING

Academic Year: 2023-2024

Program: B.Tech/MBA Tech Stream ;Data Science Year: II Semester: __IV___

Subject: Statistical Structures in Data and Inference Time: __3___ hrs (10:00am to 1:00pm)

Date: 26 / 6 / 24 . No. of Pages:__4___

Marks: __100_______

Re Examination(2022-23)

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. _1___ is compulsory.
2) Out of remaining questions, attempt any __4___ questions.
3) **In all ___5_ questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

Q1		Answer briefly:	[20]
CO- ; 2 PO- ;1 BL-1	a.	Explain the use of One-way MANOVA with a suitable example.	[5]
CO- ;1 PO- ;1 BL-1	b.	Describe the use of dummy or one-hot encoding variables in regression models.	[5]
CO- ; 2 PO- ;1 BL-1	c.	Explain how mathematically the mean of the normal distribution is calculated through maximum likelihood estimation.	[5]
CO- ;2 PO- ;2 BL-1	d.	What is Type I and Type II Error.	[5]
Q2			[20]
CO-3; PO-;1 BL-4	A	#Python Libraries import pandas as pd import numpy as np from numpy import maximum from scipy.optimize import minimize from scipy.stats import norm df=pd.DataFrame(np.random.normal(10,5,size=(100,2)),columns=list('AB'))	[10]

		The above data set(df) has two variables A and B, both the variables are Normally distributed with mean=10 and standard deviation=5. Do maximum likelihood estimation in Python to calculate mean and standard deviation of Variables A and B. Write down the python code in the answer book.	
CO-2; PO-2 BL-33	B	Explain how mathematically the parameters of the Binomial distribution is calculated through maximum likelihood estimation.	[10]
Q3			[20]
CO-3; PO-;10 BL-3	A.	Suppose we have randomly selected a stock for 100 days with up and down where: UP = 1: if a randomly selected stock is up for that day and Down = 0: if a randomly selected stock is down for that day Assuming that the stock is independent Bernoulli random variables with unknown parameter (up rate). Do maximum likelihood estimation in Python to calculate the proportion of up in the stock. Write down the python code in the answer book.	[10]
CO-1; PO-;11 BL-3	B	Suppose you are a company producing cookies. Each cookie is supposed to contain 10 grams of sugar. The cookies are produced by a machine that adds the sugar in a bowl before mixing everything. You believe the machine does not add 10 grams of sugar for each cookie. If your assumption is true, the machine needs to be fixed. You stored the level of sugar of thirty cookies by creating a distribution with 30 observations with a mean of 10 and a standard deviation of 0.05 in python. sugar=np.random.normal(10,0.05,size=(30,)) Do a testing of hypothesis in python and cross check with pre-define library, whether the level of sugar is different than the	[10]

		recipe at 0.01 level of significance. Write down the python code in the answer book.	
Q4			[20]
CO-;3 PO-;2 BL-4	A	How logistic regression is used for classification? Write a Python program to build a logistic regression model on the following data and predict the class for x_1=3.46 and x_2=59. Write down the python code in the answer book.	[10]
	B	The **cars.csv**(motor trend car road test data) dataset is consist of 32 car brands and 11 attributes. Do one way ANOVA in python and find out that whether there is any difference between average displacement for different gears and am. Write down the python code in the answer book.	[10]
Q5			[20]
CO-;3 PO-;10 BL-4	A	Two operators check the same dimension on the same sample of 10 parts. Below are the results. Is there a significant operator measurement error?	[10]

Table for Q4 A:

X_1	X_2	y
3.78	71	0
2.44	65	0
2.09	47	0
0.14	84	0
1.72	74	0
1.65	57	0
4.92	63	1
4.37	44	1
4.96	81	1
4.52	59	1
3.69	78	1
5.88	58	1

Table for Q5 A:

S.No	Operator 1	Operator2
1	63	65
2	56	57
3	62	60
4	59	58
5	62	59
6	50	57
7	63	63
8	61	61
9	56	58
10	63	64

<table>
<tr><td></td><td></td><td>Test at the 5% significance level. Do a testing of hypothesis in python and cross check with pre-define library Write down the python code in the answer book.
(hint: Pair t test)</td><td></td></tr>
<tr><td></td><td>B</td><td>A large hospital hires most of its nurses from the two major universities in the area. Over the last year, they have been giving a test to the newly graduated nurses entering the hospital to determine which school, if either, seems to educate its nurses better. Based on the following scores, help the personnel office of the hospital determine whether the schools differ in quality. Use the Mann–Whitney U test with a 10 percent level of significance. (P)
<table>
<tr><th></th><th colspan="13">Test Scores</th></tr>
<tr><td>School A</td><td>97</td><td>69</td><td>73</td><td>84</td><td>76</td><td>92</td><td>90</td><td>88</td><td>84</td><td>87</td><td>93</td><td></td><td></td></tr>
<tr><td>School B</td><td>88</td><td>99</td><td>65</td><td>69</td><td>97</td><td>84</td><td>85</td><td>89</td><td>91</td><td>90</td><td>87</td><td>91</td><td>72</td></tr>
</table></td><td>[10]</td></tr>
<tr><td>Q6</td><td></td><td></td><td>[20]</td></tr>
<tr><td>CO-;3
PO-;12
BL-4</td><td>A</td><td>The cars.csv(motor trend car road test) dataset is used which consist of 32 car brands and 11 attributes If there is any important difference, in mpg, displacement(disp) and wt, between the different gear, am and cyl or not. Perform MANOVA test in python to test the hypothesis. Write down the python code in the answer book.</td><td>[10]</td></tr>
<tr><td>CO-;1
PO-;2
BL-1</td><td>B</td><td>How qualitative predictors are incorporated into the regression model? Explain with an example.</td><td>[10]</td></tr>
<tr><td>Q7</td><td></td><td></td><td>[20]</td></tr>
<tr><td>CO-;3
PO-;10
BL-4</td><td>A</td><td>Given the following set of data:
<table>
<tr><td>x</td><td>2</td><td>4</td><td>6</td><td>8</td></tr>
<tr><td>y</td><td>3</td><td>7</td><td>5</td><td>10</td></tr>
</table>
Develop the estimating equation (regression equation) that best describes the data. (hint: Calculate parameters).</td><td>[10]</td></tr>
<tr><td>CO-;1
PO-;2
BL-1</td><td>B</td><td>How ridge regression is used to shrink estimates of the coefficients.</td><td>[10]</td></tr>
</table>